

CS/SSW 555 Agile Methods

Cyber Guru Crew (Team 17)

Sprint 3

4/09/2024

Project Background Information

Selected Project Option Number: Option 3

Main Objective of Project:

Develop a mobile application that uses AI to detect lies through voice stress, facial expressions, and response timing analysis. This application is aimed at professional and personal use, as well as it promising to enhance truth verification processes with user-friendly features.

Team Members and Roles:

Name	Role
Octavio Morales	Software Developer
Brandon Lopez	Machine Learning Analyst
Bhagawat Chapagain	Software Developer
Edmund Yuen	Machine Learning Analyst
Yunfei Tang	UX/UI Designer

List & Description of Each Sprint:

Sprint #	Objective
Sprint 1	Project onboarding and setup: - Define database schema and set up the database - Choose and set up testing framework - Basic data collection for initial model training
Sprint 2	UI/UX development - Design user interface for the app - Implement user interactions and flow
Sprint 3	Machine Learning Model development: - Select appropriate machine learning model - Train the model with collected data - Evaluate the performance of the model
Sprint 4	Machine Learning Model development:

	• Fine-tune the machine learning model • Integrate the UI components with the machine learning model
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Project Links:

Type	URL
GitHub/GitLab/Etc	https://github.com/octavio-morales1/CS555Team17.git
Jira	https://stevens-team17.atlassian.net/jira/software/projects/KAN/boards/1#
Live Demo (if applicable)	N/A

Technology Stack Used:

Type	Technology
Data Preprocessing	MatLab
UI/UX	PyQt5 (Python GUI)
Machine Learning	Kaggle
Database	MongoDB

Sprint Progress Update: Current Status

Summary of Work Completed

It's clear that our group is behind compared to others in our project. We thought we were making good progress on the lie detector, but the professor sees us as just starting. This means we need to quickly reassess our plan and work more efficiently. We'll have to look at how we're doing things, figure out where we can improve, and set clear goals. It's also important to communicate better within our team and maybe get advice from the professor or other groups (when permitted). This way, we can catch up and meet the project's expectations more effectively. Luckily, our group has believed that we have achieved such progress, putting us at par with the other groups.

Screenshot of Jira Backlog

Projects / CS 555 Team 17 Option 3

Backlog

Search: OM Epic ▾ Type ▾

☐ **Sprint 3** 31 Mar – 9 Apr (5 issues) 0 0 5 Complete sprint ...

Machine Learning Model development: Select appropriate machine learning model Train the model with collected data Evaluate the performance of t...

☒ KAN-11 Select appropriate machine learning model	DONE ▾	=
☒ KAN-12 Train the model with collected data	DONE ▾	=
☒ KAN-13 Evaluate the performance of the model	DONE ▾	=
☒ KAN-20 Populate UI/UX with relevant details	DONE ▾	=
☒ KAN-24 Fix Progress On Application [Complete REVAMP]	DONE ▾	=

+ Create issue

☐ **Sprint 1** 20 Feb – 5 Mar (3 issues) 0 0 3 Start sprint ...

Project onboarding and setup: Define database schema and set up the database Choose and set up testing framework Basic data collection for initial ...

☒ KAN-6 Define database schema and set up the databas...	DONE ▾	=
☒ KAN-7 Choose and set up testing framework	DONE ▾	=
☒ KAN-8 Basic data collection for initial model training	DONE ▾	=

+ Create issue

3 issues

☐ **Sprint 4** 5 Apr – 19 Apr (5 issues) 2 3 0 Start sprint ...

☒ KAN-23 Utilize Kaggle to train voice models	ML TRAINING ▾	=
☒ KAN-22 Implement MongoDB for secure data storage	DATA COLLECT	=

Screenshot of Burndown Chart



Video of Project at Current State:

It does not need to be one edited video. If there are multiple videos, please link all of them below:

Public Video URL (Google Drive, Loom, Etc)
No video present as this is not the current stage for visual/audio results.

Sprint Progress Update: Current Blockers

Although significant progress was made, our group is keeping the mentality of of being behind as there is much work to do. As we approach Sprint 4, it is crucial that we stay on top of our work as there are 6 KANs to do.

Looking Ahead: Next Sprint

- Finalize the secure storage solution for our data to ensure data integrity and support efficient retrieval. (KAN-22: Implement MongoDB for secure data storage)
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- Clarify and establish the database structure to lay the foundation for data management and application functionality. (KAN-18: Define database schema)
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- Refine the machine learning model's parameters and algorithms to enhance its predictive accuracy and performance. (KAN-14: Fine-tune the machine learning model)
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- Seamlessly integrate the user interface with the backend machine learning processes to provide a fluid and responsive user experience. (KAN-15: Integrate the UI components with the machine learning backend)

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